

30 Days SQL Micro Course

Day 2 – Types of Commands in SQL(DDL)

SELECT Command (DQL)

The SELECT command is used to retrieve (fetch/display) data from one or more tables.

It is the most commonly used SQL command.

1. Select All Columns

Syntax

```
SELECT * FROM table_name;
```

Example

```
SELECT * FROM Students;
```

Output

StudentID	Name	Age
-----------	------	-----

1	Amit	20
---	------	----

2	Neha	21
---	------	----

* means all columns.

2. Select Specific Columns

Syntax

```
SELECT column1, column2  
FROM table_name;
```

Example

```
SELECT Name, Age  
FROM Students;
```

Output

Name Age

Amit 20

Neha 21

3. Select with WHERE

Used to filter records.

Syntax

```
SELECT column_name  
FROM table_name  
WHERE condition;
```

Example

```
SELECT *  
FROM Students  
WHERE Age > 20;
```

Output

```
StudentID Name Age  
2          Neha 21
```

4. Select Distinct Values

Removes duplicate values.

Syntax

```
SELECT DISTINCT column_name  
FROM table_name;
```

Example

```
SELECT DISTINCT Age  
FROM Students;
```

5. Select with ORDER BY

Sorts data.

Syntax

```
SELECT *  
FROM table_name  
ORDER BY column_name;
```

Example

```
SELECT *  
FROM Students  
ORDER BY Name;
```

Descending Order

```
SELECT *  
FROM Students  
ORDER BY Age DESC;
```

Complete Example

Create Table

```
CREATE TABLE Students (  
    StudentID INT,  
    Name VARCHAR(50),  
    Age INT  
);
```

Insert Data

```
INSERT INTO Students VALUES (1, 'Amit', 20);  
INSERT INTO Students VALUES (2, 'Neha', 21);  
INSERT INTO Students VALUES (3, 'Rahul', 19);
```

Retrieve Data

```
SELECT * FROM Students;
```

Output

StudentID	Name	Age
-----------	------	-----

1	Amit	20
---	------	----

2	Neha	21
---	------	----

3	Rahul	19
---	-------	----

Common Interview Questions

What is **SELECT** used for?

Answer: **SELECT** is used to retrieve data from a database table.

What does ***** mean in **SELECT**?

Answer: It means all columns.

SELECT * FROM Students;

How do you retrieve only specific columns?

**SELECT Name, Age
FROM Students;**

Is **SELECT** a DML or DQL command?

Answer: Officially DQL (Data Query Language), though many books include it under DML.

Quick Summary

Command Purpose

SELECT Retrieve data from a table

Memory Tip

SELECT = Show/Get data from a table. ☒

Example:

**SELECT Name, Age
FROM Students
WHERE Age > 18;**

Interview Questions

Basic Questions

1. What is the **SELECT** command?

Answer: **SELECT** is used to retrieve (fetch/display) data from one or more tables.

2. What is the syntax of SELECT?

```
SELECT column_name  
FROM table_name;
```

Example:

```
SELECT Name  
FROM Students;
```

3. What does SELECT * mean?

Answer: * means all columns.

```
SELECT * FROM Students;
```

4. How do you retrieve specific columns?

```
SELECT Name, Age  
FROM Students;
```

5. Is SELECT a DML or DQL command?

Answer: Officially DQL (Data Query Language).

WHERE Clause Questions

6. What is the purpose of WHERE?

Answer: WHERE filters records based on a condition.

```
SELECT *  
FROM Students  
WHERE Age > 20;
```

7. How do you find students aged 18 or older?

```
SELECT *  
FROM Students  
WHERE Age >= 18;
```

8. How do you retrieve a specific student?

```
SELECT *  
FROM Students  
WHERE StudentID = 1;
```

9. Difference between WHERE and HAVING?

WHERE	HAVING
Filters rows	Filters groups
Used before GROUP BY	Used after GROUP BY

DISTINCT Questions

10. What is DISTINCT?

Answer: DISTINCT removes duplicate values.

```
SELECT DISTINCT Age  
FROM Students;
```

11. Why use DISTINCT?

Answer: To display unique values only.

ORDER BY Questions

12. What is ORDER BY used for?

Answer: ORDER BY sorts the result set.

```
SELECT *  
FROM Students  
ORDER BY Name;
```

13. How do you sort data in descending order?

```
SELECT *  
FROM Students  
ORDER BY Age DESC;
```

14. What is the default sorting order?

Answer: ASC (Ascending).

**SELECT *
FROM Students
ORDER BY Name ASC;**

Advanced SELECT Questions

15. How do you limit the number of rows returned?

MySQL/PostgreSQL

**SELECT *
FROM Students
LIMIT 5;**

16. How do you count rows?

**SELECT COUNT(*)
FROM Students;**

17. How do you find the maximum age?

**SELECT MAX(Age)
FROM Students;**

18. How do you find the minimum age?

**SELECT MIN(Age)
FROM Students;**

19. How do you find the average age?

**SELECT AVG(Age)
FROM Students;**

20. How do you find the total age?

```
SELECT SUM(Age)
FROM Students;
```

Pattern Matching Questions

21. What is LIKE?

Answer: LIKE searches for patterns.

```
SELECT *
FROM Students
WHERE Name LIKE 'A%';
```

Names starting with A.

22. What does % mean?

Answer: % represents zero or more characters.

```
SELECT *
FROM Students
WHERE Name LIKE '%a';
```

Names ending with "a".

23. What does _ mean?

Answer: _ represents exactly one character.

```
SELECT *
FROM Students
WHERE Name LIKE 'A___';
```

Matches names with 4 characters starting with A.

NULL Questions

24. How do you find NULL values?

```
SELECT *
FROM Students
WHERE Email IS NULL;
```

25. How do you find NOT NULL values?

**SELECT *
FROM Students
WHERE Email IS NOT NULL;**

Scenario-Based Questions

26. Display all students.

SELECT * FROM Students;

27. Display only student names.

SELECT Name FROM Students;

28. Display students older than 20.

**SELECT *
FROM Students
WHERE Age > 20;**

29. Display students sorted by age.

**SELECT *
FROM Students
ORDER BY Age;**

30. Display unique ages.

**SELECT DISTINCT Age
FROM Students;**

Frequently Asked Viva Questions

31. Which SQL command retrieves data?

Answer: SELECT

32. Which clause filters records?

Answer: WHERE

33. Which clause sorts records?

Answer: ORDER BY

34. Which keyword removes duplicates?

Answer: DISTINCT

35. Which function counts rows?

Answer: COUNT()

36. Which function finds the highest value?

Answer: MAX()

37. Which function finds the lowest value?

Answer: MIN()

38. Which function calculates the average?

Answer: AVG()

39. Which function calculates the total?

Answer: SUM()

40. Which operator is used for pattern matching?

Answer: LIKE

Top 10 Most Asked SELECT Interview Questions

- 1. What is SELECT?**
- 2. What does SELECT * mean?**
- 3. Difference between WHERE and HAVING?**
- 4. What is DISTINCT?**
- 5. What is ORDER BY?**
- 6. Difference between ASC and DESC?**
- 7. What is LIKE?**
- 8. What is COUNT(*)?**
- 9. How do you find NULL values?**
- 10. Is SELECT DML or DQL?**